

EXPERIMENT 11

QUESTIONS 36 / 48 completed

Sum of N

SUM OF N

Write a program to print the numbers from 1 to 10 using a for loop.

PROGRAM EDITOR

C++ Updated Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n;
6     cin>>n;
7     int sum=0;
8     for(int i=1;i<=n;i++){
9         sum=sum+i;
10    }
11    cout<<"sum"<<sum<<endl;
12 }
```

OUTPUT

sum55

INPUT

10

EXPERIMENT 12

QUESTIONS 36 / 48 completed

Prime

PRIME

Write a program to read in an integer and determine if it is a prime number or not using a for loop.

PROGRAM EDITOR

C++ Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n;
6     cout<<"enter number";
7     cin>>n;
8     for(int i=2;i<n;i++)
9         if(n%i==0)
10        {
11            cout<<"not prime";
12        }
13     else
14     {
15         cout<<"prime";
16     }
17     return 0;
18 }
```

OUTPUT

enter numbernot primeprime

INPUT

4

EXPERIMENT 13

QUESTIONS 36 / 48 completed

Fibonacci

FIBONACCI

Write a program to read in an integer and print the Fibonacci series up to that number using a for loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n,i,a=0,b=1,c;
6     cout<<"enter terms";
7     cin>>n;
8     for(i=1;i<=n;i++)
9     {
10         cout<<a<<" ";
11         c=a+b;
12         a=b;
13         b=c;
14     }
15     return 0;
16 }
```

OUTPUT

enter terms0 1 1 2 3 5 8 13 21 34

INPUT

10

EXPERIMENT 14

QUESTIONS 36 / 48 completed

Sum of All

SUM OF ALL

Write a program to read in an integer and print the sum of all its digits using a while loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n;
6     int sum=0;
7     int d;
8     cout<<"enter n";
9     cin>>n;
10    while(n>0)
11    {
12        d=n%10;
13        sum=sum+d;
14        n=n/10;
15    }
16    cout<<sum;
17    return 0;
18 }
```

OUTPUT

enter n12

INPUT

11433

EXPERIMENT 15

QUESTIONS 36 / 48 completed

Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n,d,r,t;
6     cout<<"enter n";
7     cin>>n;
8     t=n;
9     while(n>0)
10    {
11        d=n%10;
12        r=r*10+d;
13        n=n/10;
14    }
15    if(t==r)
16    {
17        cout<<"palindrome";
18    }
19    else
20    {
21    }
```

OUTPUT

enter npalindrome

INPUT

121

EXPERIMENT 16

QUESTIONS 36 / 48 completed

GCD

GCD

Write a program to read in two integers and compute their greatest common divisor using a do-while loop.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int a;
6     int b;
7     int r;
8     cin>>a>>b;
9     do
10    {
11        r=a%b;
12        a=b;
13        b=r;
14    }
15    while(b!=0);
16    cout<<"gcd"<<a;
17    return 0;
18 }
```

OUTPUT

gcd1

INPUT

10
29

EXPERIMENT 17

QUESTIONS 36 / 48 completed

Table

TABLE

Write a program to read in an integer and print the multiplication table for that number using a for loop.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 { int num;
5   cout<<"enter a number";
6   cin>>num;
7   for(int i=0;i<=10;i++)
8   {
9     cout<<num<<"* "<<i<<"="<<num*i<<endl;
10  }
11  return 0;
12 }
```

OUTPUT

5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50

INPUT

5

EXPERIMENT 18

QUESTIONS 36 / 48 completed

Factorial

FACTORIAL

Write a program to read in an integer and print the factorial of that number using a for loop.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5   int n;
6   int fact=1;
7   cout<<"enter number";
8   cin>>n;
9   for(int i=1;i<=n;i++)
10     fact=fact*i;
11   cout<<fact;
12   return 0;
13 }
```

OUTPUT

enter number120

INPUT

5

EXPERIMENT 19

QUESTIONS 36 / 48 completed

Perfect

PERFECT

Write a program to read in an integer and determine if it is a perfect number or not.

PROGRAM EDITOR C++ Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int n,sum=0;
5     cin>>n;
6     for(int i=1;i<n;i++){
7         if(n%i==0)
8         {
9             sum=sum+i;
10        }
11    }
12    if(sum==n)
13    {
14        cout<<"perfect";
15    }
16    else
17        cout<<"not perfect";
18    return 0;
19 }
20
```

OUTPUT

not perfect

INPUT

10

EXPERIMENT 20

QUESTIONS 36 / 48 completed

Strong

STRONG

Write a program to read in an integer and determine if it is a strong number or not.

PROGRAM EDITOR C++ Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int n,temp,d,sum=0;
5     cin>>n;
6     temp=n;
7     while(temp>0){
8         d=temp% 10;
9         int fac=1;
10        for(int i=1;i<=d;i++){
11            fac=fac*i;
12        }
13        sum=sum+fac;
14        temp=temp/10;
15    }
16    if(sum==n)
17        cout<<"strong";
18    else
19        cout<<"not strong";
20    return 0;
21 }
```

OUTPUT

strong

INPUT

145